OMIDAX

POWERED BY

DIBANET

A Sovereign Critical-Infrastructure Security Framework for Southern Africa

Cognitive anchors: Tengen's Barrier (Jujutsu Kaisen) · The Trap of King Midas
Grounded in formal cryptography, deterministic control, and lawful attribution

Research Paper and Strategic Framework — Edition v2.0

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Abstract

OMIDAX powered by DIBANET is a sovereign critical-infrastructure security framework conceived in Gaborone, Botswana. It unifies four defensive capabilities that exist separately across today's vendors and have never been combined for the Southern African context: cryptographic per-request identity grounded in elliptic-curve mathematics; a tamper-evident Merkle-chain audit and recovery spine; deterministic, rule-governed threat response in which artificial intelligence advises but never decides; and a containment-and-attribution layer that turns a successful intrusion into lawful, prosecution-ready evidence rather than a loss.

The framework is designed to run **local-first** — the evidence path works when connectivity is slow, expensive, or inappropriate for the data — while offering a **sovereignty-preserving SaaS control plane** for fleet management, policy, and intelligence. It is engineered in C++ to the NASA Power of Ten and MISRA C++ standards so that it can enter the regulated-procurement conversations closed to Python- and Java-based competitors.

OMIDAX is the layered barrier that stops the overwhelming majority of threats before they matter. DIBANET is what happens to the rare adversary who breaches it: not an escape into the real system, and not an illegal counter-strike, but entry into a fully controlled, fully recorded environment that wastes the attacker's own effort and produces the evidence that holds them accountable through law.

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1 · Conceptual Foundation — The Cognitive Anchors

Every significant architecture begins with a clear mental model. For this framework, two anchors make a deeply layered system navigable and memorable to its creator and to any technical reader who knows both domains. They are retained throughout not as decoration but as functional cognitive scaffolding.

1.1 Tengen's Barrier — the model for OMIDAX

Master Tengen's barrier in Jujutsu Kaisen is not a single wall. It is dozens of nested techniques, each serving a distinct defensive purpose, operating as one autonomous system that has run for centuries without moment-to-moment attention. Its most important properties map precisely onto OMIDAX:

- **Autonomy** — a barrier that needs constant manual tending is vulnerable whenever the tender is distracted. OMIDAX's routine operation is self-sustaining.
- **Continuous recognition** — the barrier reads an authorised entity's signature *continuously*, at every layer, not once at the gate. This is OMIDAX's continuous per-request cryptographic authentication: not a session token checked at login, but a signature verified on every request.
- **The single known breach** — Kenjaku penetrated the barrier not by overpowering it but by exploiting a partially authorised credential. This is the insider / stolen-credential attack — technically, an advanced persistent threat with credential compromise — which is how virtually

every major real breach actually happens. OMIDAX's continuous behavioural authentication and threshold authorisation are built specifically to defeat the Kenjaku attack.

1.2 The Trap of Midas — the model for DIBANET (reframed)

The myth of Midas is usually read as a parable about greed. The deeper reading is about **mechanism**: Midas was not simply punished; he was given exactly what he reached for, in a form that made his own reaching the instrument of his undoing. The gold was not the curse. Reaching for it was.

This is the philosophical foundation of DIBANET — and in v2 it is important to state precisely what the mechanism *is* and *is not*.

When an attacker breaches all of OMIDAX's layers, they believe they have reached the inner sanctum. They see data. They see access. They reach for it. That reaching is the moment DIBANET activates. From then on, **the attacker operates inside a fabricated environment that Astartis builds, owns, and records in full.** Every action they take is against Astartis's own systems and data — and so every action is lawful for Astartis to observe, record, and shape.

What the attacker experiences as progress is the steady accumulation of evidence against themselves. The fabricated data they value is worthless and wastes their time. The documents they take carry **canary markers** — a long-established, lawful attribution technique — that report when and where the stolen file is later opened. Their techniques, tools, and objectives are recorded in forensic detail. They reached for the gold; the gold was a record that leads back to them.

What DIBANET does not do, by design: it does not reach into, damage, disable, or gain unauthorised access to the attacker's systems; it does not deploy anything that executes to harm the attacker's infrastructure; it does not "strike back." Those actions are illegal in every market this framework targets and are incompatible with the certifications the framework pursues. DIBANET's power comes entirely from **information asymmetry maintained inside Astartis's own controlled space, plus lawful evidence delivered to authorities who act.** The Midas mechanism is preserved — the attacker's own effort consumes them — but it is executed within the law.

1.3 Why the anchors are retained

The framework's creator thinks in cross-domain patterns and associative anchors. These references are the map that keeps a complex multi-year architecture navigable. Their presence does not diminish the technical depth of this work. It protects it.

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2 · System Architecture — A Unified Whole

OMIDAX and DIBANET are two phases of one system that processes every entity interacting with a protected environment through a funnel with two outcomes: a legitimate, authorised entity passes smoothly, or a threat is stopped at the OMIDAX layers — and in the rare case of a full breach, contained and attributed by DIBANET.

The guiding principle is **information asymmetry held in the defender's favour at every stage**, achieved lawfully: OMIDAX learns everything about what approaches it while revealing nothing about itself; DIBANET, once a breach occurs, records everything the attacker does *inside Astartis's own fabricated environment*.

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INCOMING ENTITY — threat or legitimate request
        |
OMIDAX L1 · Perimeter Intelligence   (observe, profile, risk-score — no action
revealed)
OMIDAX L2 · Cryptographic Identity   (ECC per-request signatures — no stored secret to
steal)
OMIDAX L3 · Merkle Audit + Adaptive Response (tamper-evident log; deterministic
escalation)
OMIDAX L4 · Zero-Trust Crown-Jewel Lockdown  (microsegmentation, JIT access, threshold
authz)
        |  ~99% of threats stopped within OMIDAX
———————— BREACH THRESHOLD ————————
        |  the rare full breach → DIBANET (inside Astartis's own controlled
environment)
DIBANET L1 · Fabricated Environment  (indistinguishable replica; every action
recorded)
DIBANET L2 · Value & Decision Dilution (worthless fabricated data wastes attacker
effort)
DIBANET L3 · Attribution Telemetry   (canary markers; behavioural & tool
fingerprinting)
DIBANET L4 · Evidence & Handover     (prosecution-ready dossier → lawful authorities)
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OUTCOME — attacker's effort wasted; lawful attribution package delivered to
authorities
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Colour of the model: OMIDAX is the sovereign barrier; DIBANET is the lawful trap. Neither leaves Astartis's own boundary; neither touches the attacker's systems.

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3 · OMIDAX — The Sovereign Barrier

OMIDAX takes its name from DIMO — a figure from Setswana folklore representing a consuming, predatory force — reversed to OMID and combined with AX for *execute*: the predator turned protector. It is not a firewall (which operates on packet rules), not a SIEM (which collects logs without enforcing), and not endpoint detection (which acts after access is granted). It is a continuously operating, cryptographically grounded defensive system that scales its response from silent observation to full lockdown based on continuously assessed threat level.

3.1 Layer One — Perimeter Intelligence

The outermost layer does not defend; it watches, so as not to reveal its own structure to anything that approaches. Before any entity reaches a meaningful barrier, Layer One has built a threat model of it through four mechanisms:

- **Continuous traffic analysis** — learning the rhythms of legitimate traffic (a banking API client's request cadence, a PMU telemetry stream's polling intervals, a satellite ground-station's orbital timing windows) and flagging deviation as a *signal carried forward*, not a verdict.
- **Threat-intelligence cross-reference** — the lowest-cost, highest-return filter: checking entities against known-malicious infrastructure before spending any defensive resource.
- **Reputation and geolocation risk scoring** — a probabilistic profile that sets the *intensity of scrutiny* downstream, never an automatic block (legitimate users do operate from high-risk origins).
- **Behavioural pre-profiling** — a continuously updated model of how the entity navigates, carried through every subsequent layer.

3.2 Layer Two — Cryptographic Identity

Every password-, token-, and certificate-based system shares one structural flaw: authentication depends on a secret stored *somewhere*, and anything stored can be stolen — the Kenjaku attack in technical terms. OMIDAX Layer Two removes stored authentication secrets entirely.

- **ECC key-pair identity (secp256k1)** — every authorised entity holds an asymmetric key pair; the private key is never transmitted, never stored on an Astartis server, never shared. The public key is registered for verification. Authentication is challenge–response: Astartis issues a fresh nonce, the entity signs it, Astartis verifies against the registered public key. No secret crosses the network.
- **Continuous per-request authentication** — there is no session token to steal between requests. Every API call, data access, and administrative action is independently authenticated with a fresh signature. This defeats session hijacking, token theft, and credential mimicry.
- **Threshold authorisation** — for the highest-consequence operations (e.g. a grid black-start command, a settlement finality, a nuclear maintenance trigger), a master authorisation is mathematically split into N shares; M must cooperate within a time window to produce a valid signature. No single compromised party, no sub-threshold collusion, and no external attacker with fewer than M shares can authorise the action. This is Bitcoin-style multisig applied to the most safety-critical authorisation problem in civilian infrastructure — mathematically enforced dual control with no single point of failure.

3.3 Layer Three — Merkle Audit and Adaptive Response

Layer Three serves two functions simultaneously.

Tamper-evident audit. Every event is recorded as a log entry; each entry is SHA-256 hashed and chained into the next, so altering any historical entry invalidates every subsequent hash and is immediately detectable with no trusted third party. A regulator with access to the log can verify in seconds that nothing was modified since a published root hash. This satisfies the evidentiary requirements across the targeted verticals (IAEA safeguards reporting, PCI DSS access-log integrity, DO-178C traceability).

v2 engineering requirement: the audit chain must be **persisted to append-only storage with signed checkpoints**, not held in memory. The regulatory pitch depends on durability and external anchoring being real.

Deterministic adaptive response. Response scales with threat level through explicit, enumerable rules — not probabilistic inference:

Threat level	Triggers	Deterministic response
Low	Scan signatures, bot patterns	Silent logging, behavioural note, mild rate-limit
Medium	Credential-stuffing patterns, probe sequences	Aggressive rate-limit, challenge, enhanced logging, team notification
High	Active exploitation, authenticated-session anomaly, insider deviation	Session termination, account lockdown, incident response initiated
Critical	Coordinated multi-vector / nation-state indicators, inner-layer breach imminent	Full lockdown, non-essential access suspended, executive + lawful-authority notification, DIBANET armed

3.4 Layer Four — Zero-Trust Crown-Jewel Lockdown

The innermost layer protects data, signing infrastructure, control interfaces, and threshold key shares. No entity, however well authenticated at the outer layers, is trusted by default here.

- **Microsegmentation** — the smallest practical access domains, each with independent access control. Authentication proves *who*; it never auto-grants *access to what*.
- **Just-in-time access** — elevated privileges never persist; they are granted for a narrow window with a documented purpose and revoked automatically. There are no permanent privileged accounts to inherit.
- **HSM-backed signing** *(research-horizon for hardware; software-HSM abstraction near-term)* — operational keys generated, used, and destroyed inside tamper-resistant hardware so that full OS, memory, and filesystem compromise cannot extract them.

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4 · DIBANET — The Lawful Trap (reframed)

DIBANET takes its name from DIBA — a Setswana word for a deep, still, dangerous pool whose calm surface hides its depth. An animal that approaches sees water and walks toward it, realising the danger only when the ground gives way. This is DIBANET's operational philosophy, expressed in the landscape where it was conceived.

DIBANET is not a passive honeypot and not an offensive weapon. It is a containment-and-attribution system that activates the instant an intruder breaches OMIDAX's innermost layer. There is no alarm the attacker can see. Their experience continues seamlessly — they believe the breach succeeded. In reality they have been moved, invisibly, into a fabricated environment that Astartis owns and records in full. Everything that follows happens **inside Astartis's boundary or through lawful evidence channels**; nothing reaches into the attacker's systems.

4.1 Layer One — The Fabricated Environment

On detecting an inner-layer breach, OMIDAX performs a silent handoff: the intruder's connection is redirected, without perceptible delay or error, into a continuously maintained, high-fidelity replica of the real system — matching its data structures, response patterns, performance timing, topology, and error formatting closely enough to resist both automated tooling and careful manual inspection. Fabricated data is generated with realistic statistical properties so that round-number balances or implausibly regular histories do not betray it. Every action the intruder takes here is recorded in forensic detail. The real system, with the real data, was never reached.

4.2 Layer Two — Value and Decision Dilution

A sophisticated attacker's most valuable asset is the quality of their decisions: what to exfiltrate, where to move, which credentials to escalate. DIBANET dilutes the *inputs* to those decisions — lawfully, because it is shaping Astartis's own fabricated environment, not manipulating the attacker's systems.

- **Data-valuation dilution** — the fabricated environment presents data engineered to look maximally valuable to this category of attacker, directing their limited time toward exfiltrating material that is completely worthless while the real high-value data remains untouched in a system they never reached. Every hour spent packaging fabricated records is an hour spent doing no harm.
- **Fabricated topology** — every path the attacker follows leads deeper into Astartis's controlled replica, revealing nothing about the real network while revealing everything about the attacker's technique. Successful "lateral movement" is movement into a smaller, more fully recorded box.
- **Planted-credential attribution** — credentials placed in the locations attackers know to look are fabricated and marked. The moment one is discovered or used *inside the fabricated environment*, the event and its context are recorded as another attribution data point. The attacker is accumulating evidence against themselves.

4.3 Layer Three — Attribution Telemetry

Having characterised the attacker through Layers One and Two, DIBANET focuses on building lawful attribution.

- **Canary markers** — fabricated data elements carry canary tokens, a long-established and lawful technique (a document or token that reports back when later opened). When exfiltrated data is opened on the attacker's infrastructure, the canary reports the context of that access. This is attribution telemetry, not a payload that acts on the attacker's machine.
- **Behavioural and tool fingerprinting** — the attacker's tooling, versions, tempo, and decision style are catalogued from their behaviour *inside the fabricated environment*. Every adaptation they make generates more intelligence about their full capability.

Boundary restated: DIBANET records what the attacker does inside Astartis's environment and what lawful canary telemetry reports. It never degrades, disables, or intrudes upon the attacker's own systems.

4.4 Layer Four — Evidence and Lawful Handover

The final layer converts accumulated intelligence into two lawful outputs:

- **Defensive hardening** — every technique observed is converted into patches and configuration changes that harden the *real* system. The intrusion becomes the most comprehensive penetration test the system has ever had, conducted by a motivated adversary, at no cost and no real risk.
- **Prosecution-ready attribution package** — a forensic dossier: microsecond-timestamped actions, tool signatures, behavioural patterns, linguistic analysis of any attacker-supplied text, canary-reported access contexts, and correlation against known threat-actor profiles. It is engineered to be **legally admissible** and is delivered to the parties with lawful authority to act — national cybersecurity and intelligence agencies for nation-state attribution; law enforcement for criminal prosecution; legal counsel for civil referral. **Astartis gathers and hands over evidence; lawful authorities act.** For a government customer facing grid-targeting nation-state operations, this lawful attribution capability — which the SADC region currently lacks — is the core value.

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5 · Performance Engineering (Research Horizon)

Status: long-term ambition, not near-term product claim.

Critical-infrastructure systems operate at enormous volumes (a major bank's COBOL-bridge API, a 5G RAN telemetry stream, a grid PMU fabric, an HFT feed). At these volumes, CPU-based cryptographic processing can create latency an attacker could exploit through volumetric pressure. FPGA acceleration — implementing ECC verification, SHA-256 Merkle computation, packet classification, and inference as parallel hardware circuits — can deliver order-of-magnitude speedups with deterministic timing that removes exploitable windows.

This is retained as a **Phase 3+ research and premium-tier objective**, targeting Intel Agilex / AMD UltraScale+ for high-performance verticals and Lattice ECP5 for cost-sensitive deployments, with the most timing-critical cores (ECC verification, SHA-256 engine, Merkle accumulator) implemented in VHDL. It is explicitly **not** part of the near-term fundable product, and should not appear in early investor demos as a working capability.

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6 · Intelligence Architecture — AI Advises, Rules Decide

The intelligence layer reflects a deliberate decision: for a system protecting a grid or a hospital, a confident-but-wrong AI output — classifying a real attack as benign — is a catastrophic failure, not a tolerable error rate. The architecture therefore concentrates pattern recognition while removing AI from the final decision.

6.1 Specialist routing (near-term, deployed on local models)

Rather than one general model, a lightweight router directs each decision to a narrow specialist — traffic-anomaly, cryptographic-pattern, behavioural-profiling, threat-attribution, and deception-orchestration roles. Specialists run on **local inference** (e.g. the project's Granite-class models), preserving the local-first, no-cloud-cost promise. In the current prototype these are a **routed catalogue of persona definitions**, not many concurrent large models — a deliberately practical design for ordinary hardware. The near-term reality and the long-term vision must not be conflated in external claims.

6.2 Deterministic authority (the non-negotiable)

Every AI output must pass a **deterministic rule layer** before any security action is taken. The rules encode explicit policy and cannot be fooled by adversarial input crafted to manipulate a learned model, because they operate on the formal structure of the decision rather than on learned patterns. A specialist may output "96% similar to known C2 beacon behaviour," but that output cannot trigger an action unless the deterministic rules are also satisfied — and, conversely, the rule layer can trigger a response from other evidence even if a specialist is adversarially suppressed. **AI is advisory; the deterministic engine is the policy authority.** This is the single most important safety property of the system and the easiest to demonstrate to a regulator.

6.3 Continuous improvement and formal verification (research horizon)

Online learning from real outcomes (rewarding confirmed detections, penalising false positives and missed detections) is retained as a research-horizon capability, as is the neurosymbolic *formal* verification layer and the open mathematical questions (game-theoretic trap optimality, topological attack signatures, information-theoretic bounds, post-quantum key distribution). These are genuine and worth pursuing; they are labelled as multi-year research, not current capability.

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7 · Engineering Standards — The Real, Near-Term Moat

The choice of language and standards is a commercial and regulatory decision, not a preference. A Python core may reach SOC 2; it will not reach IEC 61513 for nuclear instrumentation. A Java core may satisfy ISO 27001; it will not pass DO-178C for avionics.

OMIDAX and DIBANET are implemented in **C++** to the **NASA Power of Ten** and **MISRA C++:2023** standards, with CERT C++ secure-coding adherence. This is the moat that is **real today and non-replicable quickly**: it opens regulated-procurement conversations closed to Python/Java competitors, and it takes years of disciplined practice to match from the first line of production code.

Requirement	C++ with MISRA / Power-of-Ten	Python / Java / Go
Deterministic timing	Native, no GC pauses	GC pauses, JIT non-determinism
Safety-cert acceptance	Accepted for DO-178C, IEC 61513	Universally rejected in formal safety certification
Static-analysis certification	Certification-grade tooling (Coverity, Klocwork)	No equivalent formal safety standard
Memory determinism	Explicit, pre-allocated	GC-driven, unpredictable

Power-of-Ten discipline (bounded loops, no post-init dynamic allocation, two assertions per function, checked returns, warnings-as-errors) is the **foundation** of the codebase, enforced in CI, not an add-on.

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8 · Deployment Model — Sovereign Data Plane, SaaS Control Plane

This section is new in v2 and resolves the central business-model question.

The buyers this framework targets — SADC banks, BPC, SAPP, ministries — are the **most** resistant to sending security telemetry to a vendor cloud. A pure SaaS is therefore unsellable to them, and contradicts the local-first promise. The resolution is a **two-plane architecture**:

- **Sovereign data plane (on-premises, per customer).** The engine, the evidence (audit chain, WORM, recovery), enforcement, and the fabricated-environment trap all run on the customer's own infrastructure. Sensitive telemetry and evidence never leave the organisation. This honours the sovereignty promise and satisfies regulators.
- **SaaS control plane (recurring revenue).** What legitimately benefits from being central: fleet management across sites and endpoints, policy authoring and distribution, threat-intelligence feeds, software updates, dashboards, and licensing. The control plane manages; it does not ingest the customer's protected data.

This makes the multi-host capability — a central server coordinating sensors/agents across endpoints and segments — **both** the core product gap to close **and** the foundation of the SaaS business. "Fleet" in the product means managed hosts; it is distinct from the AI persona catalogue.

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9 · Application Across Critical-Infrastructure Verticals

The framework targets verticals that are underserved by the current market, willing to pay for security that meets specific regulatory requirements, and consequential when security fails. The near-term strategy is **one vertical first**, not ten at once.

Vertical	OMIDAX application	DIBANET value (reframed)	Primary standards
Banking & COBOL *(recommended first entry)*	ECC identity for COBOL-bridge API gateways, Merkle audit for compliance, threshold authz for settlement	Contain and attribute credential-theft and lateral-movement attempts against mainframe bridge interfaces	PCI DSS, SOC 2 Type II, ISO 27001
Power Grid & SAPP	Zero-trust for PMU/SCADA, grid-control traffic classification	Contain grid-destabilisation attempts; lawful attribution for SAPP interconnect attacks	IEC 61850, IEEE C37.118, NERC CIP
Telecommunications	Zero-trust for RAN telemetry and 5G slice-management APIs	Attribution for infrastructure-mapping and BGP-hijack preparation	3GPP TS 28.532, ETSI NFV
Aviation	Cryptographic integrity for flight data and MRO APIs	Attribution for ground-system intrusion attempts	DO-178C, DO-326A
Nuclear	Merkle logs for IAEA reporting, threshold authz for maintenance triggers	Containment + attribution for nation-state intrusion against plant systems	IEC 61513, IEC 60880, IAEA SSG-39
Healthcare	HIPAA-aligned Merkle audit, ECC identity for DICOM/FHIR	Containment + attribution for patient-record exfiltration	HIPAA, IEC 62304, HL7 FHIR
Defence / Satellite / Fibre / HFT	Threshold authz, zero-trust control-plane protection	Lawful attribution packages for state-sponsored and criminal attacks	MIL-STD-1553, CCSDS 201.0-B, MEF LSO, MiFID II

9.1 Banking & COBOL — the first commercial entry point

Every major SADC bank runs COBOL mainframes at its core; the threat comes not from the mainframe but from the **API bridges** exposing mainframe data to modern and mobile applications. OMIDAX makes the bridge safe (ECC identity per API client, Merkle audit for regulatory reporting, threshold authz for settlement finality). This is a concrete, fundable first problem in Botswana.

9.2 Power Grid & SAPP — the strategic African priority

The Southern African Power Pool coordinates generation across 12 SADC countries. An attack on the SAPP interconnect's control infrastructure propagates regionally. OMIDAX's zero-trust for PMU/SCADA and threshold authz for AGC/black-start operations address this profile; DIBANET provides the **lawful attribution capability the region currently lacks** for nation-state grid-attack attempts. A SAPP contract is a 12-country reference case, not a single-country deal — the largest strategic prize, and the slowest. Pursue after a banking reference exists.

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10 · Commercialisation Strategy

Four compounding paths:

1. **Standalone enterprise security product** — direct licensing across verticals, priced by client tier.
2. **Bundled with the Advanced API Firm's COBOL-bridge APIs** — the strongest near-term move, because it solves the cold-start trust problem: the security layer earns its track record *inside* an API the client already runs and values, with no standalone security procurement required.
3. **Patent licensing** — file narrowly, on working implementations, for the three defensible, non-obvious methods: FPGA-accelerated line-rate Merkle audit logging; threshold multi-party authorisation for safety-critical control APIs; and the lawful adversarial-deception-and-attribution method. File in US, EU, South Africa, and Botswana with PCT priority. *(File when implemented, not before.)*
4. **Government / national infrastructure** — the long institutional play: BOCRA, DISS, BPC, and SAPP, building the continental reference case.

Revenue framing (scenarios tied to milestones, not headline promises):

Horizon	Stage	Scenario revenue	Gating milestone
Yrs 1–3	Working OMIDAX L1–L2 prototype, multi-host, persisted evidence, docs + IP dated	Pre-revenue	Demoable sovereign engine + one pilot LOI
Yrs 4–5	DIBANET containment/attribution prototype, SOC 2 Type I, bundled pilots	Early pilot revenue	First paying bundled client
Yrs 5–6	Patents filed, SOC 2 Type II, first standalone enterprise contracts	Growth	Certification + first patent granted
Yrs 6–7	First government contract, SADC expansion begins	Scale	BPC or Bank-of-Botswana reference
Yrs 7–10	Multi-vertical, premium (FPGA) tier, SAPP engagement	Mature	SAPP interconnect + multiple SADC governments

Headline valuation figures from v1.0 are intentionally omitted here; valuation follows from ARR and the patent portfolio, and overstated figures read as a red flag to experienced investors. State the moat as **African regulatory access + sovereignty + certification discipline**, not the feature count.

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11 · Competitive Landscape

The honest reading: capable, well-funded companies have built *pieces* of this space for years, but none combines a unified, certification-grade, deterministic defensive engine with lawful deception-and-attribution, engineered for the SADC context. The market is one of strong **partial** solutions.

Company	Core capability	Gap vs. OMIDAX + DIBANET (reframed)
Palo Alto Networks	NGFW, detection, cloud platform	No certification path for nuclear/aviation; no lawful containment-and-attribution layer; absent from SADC critical-infra
CrowdStrike	Endpoint detection/response, identity	Statistical matching without deterministic authority layer; no deception/attribution spine; no safety-critical certification
Illusive / Attivo (SentinelOne)	Enterprise deception	Deception only; no cryptographic identity, no certification, no prosecution-ready attribution package, no regional presence
IBM Security	Hybrid/mainframe enterprise security	General enterprise focus; no SADC regional context; no unified deterministic-plus-attribution design
Darktrace	AI anomaly detection	Pure neural approach (hallucination risk in decisions); no deterministic authority; no safety-critical certification
Claroty / Dragos / Nozomi	OT/ICS monitoring	Monitoring/detection only; no lawful containment-and-attribution; no certification path for nuclear/aviation

The real, non-replicable moat is not the feature stack — it is that no incumbent has SADC critical-infrastructure relationships (BOCRA, BPC, SAPP, SADC central banks), has adapted to the region's regulatory and threat context, or has commercial incentive to, given market size relative to their North American and European bases. The creator is Motswana, educated in Botswana, building with deep understanding of that context. That alignment between creator, product, and market is the advantage an external competitor cannot copy. **The post-breach gap that matters is lawful containment-and-attribution — not strike-back — and that is the category OMIDAX + DIBANET is built to fill.**

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12 · Development Roadmap

Built on an honest accounting of the creator's position: 18, employed at a family IT-tendering firm, entering a Computer Systems Engineering degree, with strong conceptual clarity and an early prototype. The path is disciplined and sequential.

Phase	Timeline	Technical milestones	Commercial milestones
0 — Foundation	Now → university entry	Reframe this paper (done); fundamentals; ECC/SHA-256/Merkle study; **persist the audit chain, make policy file-driven**	Timestamp + legally date all IP; GitHub + technical presence
1 — Degree Yrs 1–2	Yrs 1–2	C++ to Power-of-Ten; OpenSSL ECC identity; Merkle prototype; **multi-host sensor→server architecture (data plane + SaaS control plane v0)**	First technical paper; first BOCRA/BPC/bank IT contacts
2 — Degree Yrs 3–4	Yrs 3–4	Full OMIDAX L1–L4 prototype; DIBANET fabricated-environment + attribution prototype (no strike-back); first external pen-test	Patent applications on implemented methods; SOC 2 Type I gap analysis
3 — Post-grad Yr 1	Yr 5	Research-horizon integration begins (FPGA, formal verification); hardening	First bundled pilot with COBOL-bridge APIs; SOC 2 Type II initiated; first government meeting
4 — Early commercial	Yrs 6–7	IEC 61850 (grid) / DO-178C traceability (aviation) work; QKD prototype	First standalone enterprise contracts; first SADC government contract; patent licensing talks
5 — Full commercial	Yrs 8–10	Ten-vertical capability; FPGA premium tier; post-quantum option	SAPP engagement; multiple SADC governments; patent royalties

The single most important near-term action — before scaling code — is documentation and legal dating of the IP, plus the two engineering moves that convert the demo into a product: **persisted evidence** and **multi-host architecture**. Everything else builds on those.

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13 · Conclusion

OMIDAX powered by DIBANET addresses the most consequential unsolved problem in enterprise security: what happens after a sophisticated attacker breaches the perimeter. The market's conventional answer — detect, isolate, investigate — is reactive and imposes no cost on the attacker. OMIDAX answers the perimeter with four layers of autonomous, cryptographically grounded, deterministic defence. DIBANET answers the post-breach moment not with detection-and-remediation, and **not** with an illegal counter-strike, but with lawful containment: the attacker falls into a fully controlled, fully recorded environment, wastes their own effort on worthless fabricated data, and produces a prosecution-ready attribution package delivered to the authorities with the power to act. The attacker reached for the gold; the gold was a record that leads back to them — collected, and handed to the law.

The commercial opportunity is genuine, large, and structurally open: a combined global market in the hundreds of billions, with tens of billions of addressable SADC opportunity that major Western vendors have neglected. The decisive, non-replicable advantage is the alignment between a

Motswana creator, a sovereignty-preserving product, and an underserved regional market with specific regulatory needs.

The distance to a deployed, certified, commercially operational system is measured in years, and this edition is honest about that: a fundable near-term wedge (the deterministic sovereign engine, made multi-host, with real evidence and configurable policy, entering through banking), and a long-horizon research ambition (FPGA, formal verification, post-quantum, and the open mathematics) behind it. The direction is correct, the architecture is sound and now **lawful and certifiable**, the market gap is real, and the work is disciplined.

Everything that will exist starts exactly here — within the law, and built to last.

Appendix A — Summary of Changes from v1.0

This strategic edition preserves the vision, mythology, market thesis, and engineering ambition of the original paper. Four changes make the framework legal to sell, possible to certify, and fundable.

This edition keeps the vision, the mythology, the market thesis, and the engineering ambition of v1.0 intact. It changes four things so the framework is **legal to sell, possible to certify, and fundable by serious investors**. Review these decisions before the body:

1. **DIBANET is reframed from "active counterattack" to "containment, deception, attribution, and evidence."** v1.0 described beacon payloads, resource exhaustion, tool-blinding, and "complete attacker destruction" aimed at the attacker's own infrastructure. That is **hack-back** — illegal under the US CFAA, EU law, and Botswana's Cybercrime and Computer Related Crimes Act. No bank, utility, or ministry can lawfully buy it; no investor or acquirer will touch a company whose core differentiator is a crime; and it cannot pass the DO-178C / IEC 61513 certification this paper leans on as its moat. **The reframe loses nothing commercially and keeps the Midas story whole:** the attacker still defeats themselves. The difference is *where* it happens. Everything in reframed DIBANET occurs **inside Astartis's own environment or through lawful evidence channels** — never by reaching into the attacker's systems. The attacker exhausts *their own* time and resources operating inside a space Astartis fully controls and records, and the output is a **prosecution-ready attribution package handed to authorities who act**. For a government buyer facing nation-state grid attacks, lawful attribution is *more* valuable than an illegal strike-back they could never deploy.
2. **The architecture is explicitly split into a sovereign on-prem data plane and a SaaS control plane.** v1.0's promise is local-first, data-never-leaves. A pure SaaS contradicts that and is unsellable to the exact buyers targeted. v2 resolves this: **evidence and enforcement stay on the customer's premises; management, policy distribution, threat intelligence, and licensing are the recurring-revenue SaaS layer.** Sovereignty preserved, SaaS business model intact.
3. **Capabilities are staged into "near-term product" and "research horizon."** FPGA acceleration, quantum key distribution, neurosymbolic formal verification, and the open mathematical proofs are retained as genuine long-term ambition — but clearly labelled as Phase 3+ research, not current capability. The fundable wedge is the deterministic defensive engine, made multi-host, with real persisted evidence and configurable policy.

4. **Claims are tightened to what is defensible.** Valuations are framed as scenarios tied to milestones rather than headline figures; "no competitor combines this" is reframed around the real, non-replicable moat (African regulatory access + sovereignty), not the feature stack.

The remainder of this document is the reframed paper.

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